

LECTURE 0: INTRODUCTION (V1)

1. INTRODUCTION

Computational algebraic geometry. That is the title of the course, and, on the one hand, that title is quite apt. There is a large and varied part of mathematics—more than I could reasonably cover, or even survey, in five lectures—devoted to computation inspired or informed by algebraic geometry. It ranges from the use of combinatorics and polyhedral geometry in toric and tropical geometry, through homological methods based on free resolutions, to numerical and homotopy methods for finding and certifying solutions to particular systems over $\mathbb{R}$ or $\mathbb{C}$, and much more.

However, having finished writing 95% of these notes—the remaining 5% being, well, what I am currently writing—the title also feels like a misnomer. These notes are not about a field separate from algebraic geometry or a niche subfield within algebraic geometry. Instead, I wish to suggest that, in many ways, computation is the spiritual heart of algebraic geometry. All these notes attempt to do is take that viewpoint seriously, with the modern tools at our disposal in mind.

The claim that computation is the spiritual heart of algebraic geometry may seem indefensibly bold. After all, over the past seventy-five years, the field has become known to many outsiders for its abstract and formidable machinery: schemes and sheaves, a plethora of cohomology theories, stacks and derived categories, and, more recently, even more exotic things such as shtukas, homotopical algebra, ∞ -categories, etc. So let me take a moment to defend the claim with two arguments.

1.1. Examples in Algebraic Geometry. Algebraic geometry seeks to study the shapes cut out by systems of polynomial equations. Such shapes lurk in a rather large and dark forest. The central questions that have driven the modern field ask us to chart that forest: Which shapes exist? How do they vary? Which features are typical? Which apparent possibilities never occur? Faced with a forest this large, it may seem absurd that inspecting a single tree could tell us much. Yet, remarkably, it often can.

An enormous number of these questions—including some of the most celebrated ones—come down, in the end, to a single humble task: *compute something about a single concrete example*. We must not merely describe such an example abstractly, but build it and prove something true about it. What makes this possible is a piece of true magic worth stating up front. For many of the properties we care about, the locus of members having the property is *open* in the relevant parameter space: if one member has the property, then so does every sufficiently nearby member. Provided that parameter space is irreducible, any nonempty open locus is dense, so the property holds for what algebraic geometers call a “general” member. Showing that a property is open, while not always immediate, often seems to come down to finding the right definitions. Of course, the empty set is

open! Thus, once openness is known, to prove that a general object has the nice property we care about—and to be certain that we are not proving a vacuous theorem about nothing—it is enough to construct a specific example with that property. In this sense, one well-chosen example becomes a theorem about a dense open subset of the family. In fact, I do not think I am lying when I say that one could win a Fields Medal simply by constructing the right example.

Once you know this pattern, it is impossible to miss. To convince you that I am not lying—and perhaps to argue from authority by appealing to papers that have appeared in so-called fancy journals—let me briefly list a few of my favorite instances: Brill–Noether theory, Green’s conjecture for a general curve, the Maximal Rank Conjecture, and interpolation problems. You are not expected to know these results yet; the common pattern is the point.

1.2. Abstraction in Algebraic Geometry. The framing of the previous paragraph deserves to be challenged rather than answered. It supposes that the machinery—sheaves, cohomology, resolutions, derived categories—sits at the opposite end of the subject from computation, so that the best one could hope to show is some historical link between the two. However, the historical record suggests this is quite far from the case.

Take the machinery at its source. Hilbert’s theorem that rings of invariants are finitely generated produced no generators, and Paul Gordan—who had spent his career producing generators by hand—is supposed to have complained, “*Das ist keine Mathematik. Das ist Theologie!*” The anecdote is too tidy to be true, and, more importantly, it stops far too early. Pressed on precisely the question of effectivity, Hilbert kept going: the Basis Theorem, the Syzygy Theorem, and the Hilbert function all come out of the effort to get his hands on the invariants he had shown to exist. Faced with a computation he could not finish, he invented graded free resolutions. That is not a retreat from calculation into abstraction. It is what the calculation turned into.

Noether then did to Hilbert what Hilbert had done to Gordan. The Basis Theorem, as Hilbert proved it, is a statement about polynomial rings over a field, and its proof is entangled with the elimination-theoretic setting it came from. Noether asked what the argument had actually used. The answer was: very little. Every ideal finitely generated, or equivalently the ascending chain condition, and nothing else – so that the theorem is better stated as the assertion that this single property passes from R to $R[x]$. A hypothesis extracted from one proof became the definition of a class of rings, and Hilbert’s theorem became the statement that the class is closed under adjoining a variable. The path here is the one worth naming:

calculation $\longrightarrow$ pattern $\longrightarrow$ abstraction $\longrightarrow$ better calculation,

and it is the last arrow that matters for us. Noetherianity is not a piece of bookkeeping we tolerate on the way to the interesting theorems. It is the reason Buchberger’s algorithm terminates and the reason free resolutions are finite, which is to say it is the reason that essentially every computation performed in these notes returns an answer at all. Gordan, incidentally, later gave a proof of Hilbert’s theorem whose monomial arguments read today like an anticipation of Gröbner bases; the two sides of the quarrel converged on the same object.

The Path Through These Notes. In writing these notes, I have tried—and we shall see to what extent I have succeeded—to weave together a number of threads that appear, disappear, and reappear throughout the lectures. At times, an exercise will deliberately be left hanging, a calculation left unexplained, or a theorem left unproved. This is intentional—or I did not catch it in my hurried proofreading. You will see relatively few proofs presented in the traditional theorem–proof format. Sometimes a proof lies beyond the scope of these notes and is best revisited after you have built more machinery; other times, the proof is buried in the exercises. That, too, is deliberate. If you are reading this, I hope your goal is to learn something, and, well, in my experience, no amount of my knowing something can magically make another person know it, too. (The same statement is even more true with the roles reversed.) Learning requires doing, and I hope these notes serve as a guide while you do that work for yourself. That is not to say that I intend to send you into the woods of algebraic geometry unprepared and unaccompanied. Instead, I have tried to write in the spirit of friends setting out on a hike together: I have walked enough of the trail to point toward the next turn and warn you about the loose rocks, but you still have to walk it yourself.

Concretely, the five lectures follow this route:

- **Lecture 1: Equations and Geometry.** We ask what it could possibly mean to *solve* a system of polynomial equations, a question that turns out to have at least six answers! This leads us to study algebraic varieties, ideals, coordinate rings, the operation of taking the radical of an ideal, and Hilbert’s Nullstellensatz. Along the way, our four major recurring players enter the scene: the intersection of a parabola and a circle, a plane conic, the twisted cubic, and an elliptic curve. More about them later.
- **Lecture 2: Division and Hidden Equations.** One-variable division suggests a nonlinear analogue of row reduction, so we try to imitate it. Monomial orders and multivariable division show us why an arbitrary list of generators is not good enough; initial ideals and standard monomials tell us what a better list would have to accomplish; and S -polynomials isolate the one cancellation that seems to build such a list every time we try it.
- **Lecture 3: Finding Gröbner Bases.** Was that luck? Dickson’s Lemma delivers finite Gröbner bases and, as a bonus, Hilbert’s Basis Theorem; Buchberger’s criterion turns an infinite condition into a finite test; and Buchberger’s algorithm actually constructs a basis. We then pass to projective space, where a Gröbner basis for a degree-compatible monomial order lets us homogenize an affine ideal without the false components at infinity that naively homogenizing an arbitrary generating set can happily invent.
- **Lecture 4: Questions of Rationality & Interpolation.** Here we reverse the direction of travel and start building. The Elimination Theorem recovers equations for the Zariski closure of a parametrized image; standard monomials turn interpolation on a finite set into linear algebra; and rational and birational maps make precise when two apparently different varieties carry the same rational functions—and when, as for a smooth plane cubic, no

birational parametrization by $\mathbb{P}^1$ exists. Both themes named in the title are entry points into broad research programs in which basic questions remain open.

- **Lecture 5: What an Initial Ideal Remembers.** Finally, we see that what we have been doing with Gröbner bases has an entirely different geometric meaning. We read initial ideals as special fibers of degenerations, count standard monomials using Hilbert functions and Hilbert polynomials, and watch a complicated variety move to a combinatorial model. The point set, smoothness, and components may all change along the way; we ask precisely what does not, because that is what a degeneration argument is allowed to conclude.

1.3. Using These Notes. Each lecture builds on its predecessors, and I would not encourage you to read them backwards. At the same time, you should not start at the first page and march steadily forward, moving on only after you have finished and digested every part of every exercise. That is not how these notes are meant to be used, and it is not how I find mathematics is best learned. Instead, I encourage you to try many parts of many exercises. If you stall on an exercise, move forward, do something else, and eventually come back. Let things simmer while you move on: as new ideas appear and the bigger picture comes into focus, earlier exercises that seemed daunting may become clearer.

There are also deliberately more exercises, parts, and remarks than we can cover together. Honestly, I have included far more than I expect anyone to cover and digest deeply in five lectures, in the hope that these notes will serve as a friendly introduction to the subject. Still, if you find the notes at all interesting and continue studying algebraic geometry, I like to imagine that you might, from time to time, come back to them and revisit some old friends and familiar examples in a new light.

When it comes to computation, I encourage you both to compute by hand and to compute with *Macaulay2*. Each is a valuable skill in its own right. *Macaulay2* is free, and a browser-based interface is available, so there is nothing to install before trying it. You can, of course, also download it and run it locally, something you will definitely want to do if you continue using it. For working with *Macaulay2*, I have three suggestions:

- (1) Instead of copying and pasting the snippets of code in these notes directly into a session, retype them by hand. Yes, this may be old-fashioned, but the process of typing things out often helps us remember commands, syntax, and structure more deeply than copying and pasting does. Pause over each command and ask what mathematical object it creates or computes.
- (2) Consult the documentation frequently. The documentation for *Macaulay2* is written largely by research mathematicians, many of whom work in algebraic geometry, and is a treasure trove of examples, ideas, and papers. It is generally the best way to learn the language.
- (3) If you enjoy using *Macaulay2*, consider getting involved! The *Macaulay2* is a vibrant and diverse group of researchers from numerous areas of mathematics. There are frequent

workshops dedicated to building and extending the functionality of the language, and they are just a lot of fun.

1.4. Apologia. I must end with a fairly serious warning and an apology. These notes are very, very much a first draft and an experiment. This is both the first time I have taught a course like this and the first time these notes have been used. I am certain that there are mathematical errors, typographical mistakes, idiosyncratic explanations that make sense only to me, and exercises that do not work as I had hoped. These issues and the perhaps peculiar choices scattered throughout the notes are entirely mine. The lectures were written at different times, sometimes weeks or months apart. So I apologize if the tone, voice, or tenor of the writing ever shifts in a way that feels jarring.

In the hope of correcting some of these issues and of saving any future souls who find and use these notes, I have created a Google Form through which you can report any issues you find.

<https://forms.gle/eWsQYhPhvA3edpfSA>

I would be genuinely grateful for any corrections, comments, or other feedback you might have. When submitting feedback, please keep in mind that it is much easier to act on specific, isolated comments. You are, of course, welcome to use the form to tell me that you hate everything about these notes and that they are useless, but I doubt I will know what to do with that. On the other hand, being told that I missed a sign on the third line of the fourth exercise in the second lecture should result in a speedy and satisfying resolution.

LECTURE 1: EQUATIONS AND GEOMETRY (V1)

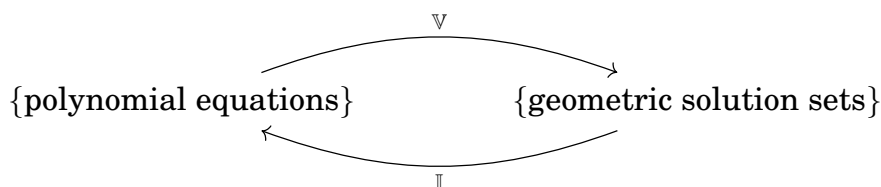
Throughout this lecture, $\mathbb{K}$ denotes a field. The algebra–geometry dictionary we shall build works over an algebraically closed field, but we will often restrict to $\mathbb{R}$ when we wish to visualize things. Further, for reasons we will explain shortly we generally do computations over $\mathbb{Q}$.

1. WHAT DOES IT MEAN TO SOLVE?

The passage between algebra and geometry is familiar to us from high school. Consider the equation

$$y = x^2,$$

the solutions to which determines a subset of the plane $\mathbb{R}^2$, namely its graph. Conversely, if we are shown a graph in $\mathbb{R}^2$, we may try to recover the equation(s) which cut it out. There is something remarkable hidden in this familiar example. Formally the parabola as a subset of $\mathbb{R}^2$ consists of infinitely many points, however, the single expression $y - x^2$ encodes all of them in one piece of finite data. Algebraic geometry takes this familiar exchange seriously:



The two directions lead to different questions. The *forward problem*, which we denote $\mathbb{V}$ for reasons that will become clear shortly, asks us to find the common solution set to a given system of polynomial equations, i.e. we are asked to solve a polynomial system. The *inverse problem*, denoted by $\mathbb{I}$ for reasons that we will see later in this lecture, asks how we can find polynomial equations that cut out a given subset, i.e. we are asked to identify the graph.

The bulk of this lecture is devoted to understanding both directions of this relationship in detail, beginning with the forward problem and then considering the inverse problem. Towards this let us consider a relatively simple system of polynomial equations, which none the less will show that there are several different questions hidden in the instruction *solve the system*.

(1) A Circle & A Parabola. Consider the following system of two polynomial equations:

$$f_1 = x^2 - y = 0, \quad f_2 = x^2 + y^2 - 1 = 0.$$

Since f_1 defines a parabola and f_2 the unit circle at the origin we are looking at the intersection of two very familiar shapes.

- (a) Substitute the first equation into the second and show that every solution satisfies

$$x^4 + x^2 - 1 = 0.$$

The roots of this new polynomial are the possible x -coordinates of the intersection. What information was eliminated in passing from two equations in x, y to one equation in x ? Here the equation $y = x^2$ reconstructs the discarded coordinate. Explain why every root lifts uniquely.

- (b) Now work over $\mathbb{R}$. Set $u = x^2$ and solve $u^2 + u - 1 = 0$. Show that exactly one of its roots is nonnegative.
- (c) Find the two real solutions of the original system and sketch the picture.
- (d) Now work over $\mathbb{C}$. Show that $x^4 + x^2 - 1$ has four distinct roots and conclude that the system has four complex solutions. You need not choose a preferred square root of each complex number: an answer of the form $(\pm\sqrt{u}, u)$ is more informative.
- (e) Subtract the first equation from the second to obtain a polynomial involving only y . Explain why this polynomial vanishes at every solution.
- (f) Compare what is recorded by the real picture, the complex solution set, and the two one-variable elimination equations obtained in parts (a) and (e). What information does each description retain, and what does it leave out?
- (g) A projection can lose information in two different ways. Over $\mathbb{C}$, every nonzero x -coordinate on the curve $y^2 = x$ has two lifts, so the discarded coordinate cannot be recovered uniquely. Now assume $\mathbb{K}$ is infinite and consider the hyperbola

$$H = \{(x, y) \in \mathbb{K}^2 \mid xy = 1\}.$$

Show that its projection to the x -axis is $\mathbb{K}^\times$. Prove that the only polynomial in $\mathbb{K}[x]$ which vanishes on this image is the zero polynomial. (Recall that a nonzero one-variable polynomial has at most as many roots as its degree.) Thus any common zero set of polynomials in $\mathbb{K}[x]$ which contains the image must be the whole line, even though the origin is not in the image. Here $y = 1/x$ does recover the discarded coordinate on the image; the failure is instead that the image cannot itself be described as a common zero set of polynomial equations. We will recover this distinction by elimination later in the lecture.

The previous exercise suggests that the instruction *solve the system* is a bit ambiguous. For a collection of polynomials $\mathcal{F} \subset \mathbb{K}[x_1, \dots, x_n]$, there are several different geometric questions one might be asking. The following table captures several more specific versions of what one might hope to do when solving a polynomial system. This one might have when actually trying to solve a system of polynomial equations

Geometric question	Algebraic form	<i>Macaulay2</i>
Does a solution exist?		
Which equations hold on every solution?		
Which consequences involve only selected coordinates?		
Is the solution set finite or positive-dimensional, and how many points occur when multiplicity is remembered?		
Does the set break into irreducible pieces?		
Which equations vanish on the image of a parametrization or on the closure of that image?		

Throughout this course we will use the word *solve* in each of these senses. Each geometric question has an algebraic counterpart, and our eventual goal is to make these algebraic questions computable. We shall frequently return to this table throughout every lecture, and please keep it in mind. The model is Gaussian elimination, which answers the linear versions of these questions.

(2) **Linear Equations - The Gold Standard.** Work over an arbitrary field $\mathbb{K}$ and consider the following system of linear polynomials:

$$\ell_1 = x + y + z - 1, \quad \ell_2 = 2x - y + z.$$

- (a) Row reduce the system $\ell_1 = \ell_2 = 0$ and write its solutions as a one-parameter family.
- (b) Show that

$$x - 2y + 1 = \ell_2 - \ell_1, \quad 3y + z - 2 = 2\ell_1 - \ell_2.$$

Use these equations to recover your parametrization from part (a).

- (c) Explain why an elementary row operation replaces a list of linear polynomials by another list generating the same ideal. Conclude, without appealing to the geometry, that row operations preserve the common zero set.
- (d) Suppose a linear system in n variables is consistent and its coefficient matrix has rank r . What is the dimension of its solution set? What comparison between the coefficient matrix and the augmented matrix detects inconsistency?
- (e) Consider instead the linear equations

$$m_1 = x + y, \quad m_2 = 2x + 2y - 1.$$

Verify the identity $1 = 2m_1 - m_2$. Explain why this identity is an algebraic certificate that the system $m_1 = m_2 = 0$ has no solutions over *any* field.

(f) Start now with the parametrized line

$$L = \{(2t - 1, t, 2 - 3t) : t \in \mathbb{K}\} \subseteq \mathbb{K}^3.$$

Find two independent affine-linear polynomials which vanish on L . This is the inverse problem for a linear set.

(g) Match parts (a)–(f) with the rows of the opening table which they address. Which row has not yet appeared in a substantial way? Which steps use only ideal-preserving combinations of equations, and which use the special echelon organization of leading variables? What nonlinear replacement is still missing?

2. FROM EQUATIONS TO IDEALS

For systems of linear equation, replacing a subset of the equations by suitable linear combinations, does not change the underlying geometry of their solution set. Further, via these operations we may eventually move our system of equations into echelon form from which the geometry is very visible. For systems of non-linear equations, we do not have an analogous notion of echelon form, but there are natural operations, analogous to linear combinations, which preserve the solution set. Namely, we may replace the care list of equations by the ideal the list of equations generates. In partocular, if $f_1, \dots, f_s$ vanish at a point $p \in \mathbb{K}^n$, then every expression of the form

$$h_1 f_1 + \dots + h_s f_s$$

for $h_1, \dots, h_s \in \mathbb{K}[x_1, \dots, x_n]$ also vanishes at p . At a moment to ponder why the converse is also true. Thus passing from a list of equations to its ideal records all consequences obtained by these elementary algebraic operations without changing the common zero set.

Definition 1. Let $\mathcal{F} \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an arbitrary subset. The *vanishing set* of $\mathcal{F}$ over $\mathbb{K}$ is

$$\mathbb{V}_{\mathbb{K}}(\mathcal{F}) = \{p \in \mathbb{K}^n \mid f(p) = 0 \text{ for every } f \in \mathcal{F}\}.$$

When the ground field $\mathbb{K}$ is clear, we simply write $\mathbb{V}(\mathcal{F})$. A subset of $\mathbb{K}^n$ of the form $\mathbb{V}(\mathcal{F})$ is called an *affine algebraic set* or an *affine algebraic variety*.

Out of economy, or perhaps laziness, we often drop adjectives and simply say *algebraic set*, *algebraic variety*, or just *variety* when it seems little confusion will arise. As a slight warning, there are dueling notations throughout the algebraic geometry over the use of the word variety. Some authors reserve the term algebraic variety for an algebraic set, which is also irreducible. (A topological property we shall see later in this exercise.) Others adopt the slightly broader definition we have taken where algebraic set and variety are synonymous and we shall stress the adjective irreducible when appropriate.

Notice the collection $\mathcal{F}$ in the above definition is allowed to be infinite. Nothing, so far, rules out an algebraic set which genuinely requires infinitely many equations. One of the central results of these lectures, Hilbert's Basis Theorem, which we will prove in Lecture 3, says that this cannot occur: every affine algebraic variety can be cut out by finitely many equations.

The notation $\mathbb{V}(\mathcal{F})$ admits two useful mnemonics: $\mathbb{V}$ for the common *vanishing* locus and $\mathbb{V}$ for the *variety* it defines. The historical origin of the this notation is a bit mysterious, however, the mathematical word *variety* comes to use from various translations of Riemann's *Mannigfaltigkeit*. While in modern usage this word translates as manifold, Riemann's use of the word was much more general than the current usage of a smooth or even topological manifold. The German adjective *mannigfaltig*, which is a root of *Mannigfaltigkeit*, roughly translates as plural, many, or perhaps various, which seems to be in roughly line with Riemann's work. As such Italian mathematicians translated *Mannigfaltigkeit* as *varietà*, and many French mathematicians translated it as *variété*. Note other French mathematicians translated it as *multiplicité* or *continuum*. When this word began solely referring to algebraic solution sets is unclear.

(3) From Equations to Ideals. Let $\mathcal{F} \subseteq \mathbb{K}[x_1, \dots, x_n]$ and let $I, J \subseteq \mathbb{K}[x_1, \dots, x_n]$ be ideals.

- (a) Prove that $\mathbb{V}(\mathcal{F}) = \mathbb{V}(\langle \mathcal{F} \rangle)$. Thus the common zero set remembers the ideal generated by the equations, not merely the displayed list.
- (b) Prove that $\mathbb{V}$ reverses inclusions: if $I \subseteq J$, then $\mathbb{V}(J) \subseteq \mathbb{V}(I)$.
- (c) Describe the following varieties, and when possible, draw the corresponding picture. Over $\mathbb{C}$, a literal sketch is likely difficult, so you may need to be creative.

$$\begin{aligned} \mathbb{V}_{\mathbb{R}}(x^2 + y^2 - 1), \quad \mathbb{V}_{\mathbb{R}}(x^2 - y^2), \quad \mathbb{V}_{\mathbb{R}}(x^2 + y^2), \\ \mathbb{V}_{\mathbb{C}}(x^2 + y^2 - 1), \quad \mathbb{V}_{\mathbb{C}}(x^2 - y^2), \quad \mathbb{V}_{\mathbb{C}}(x^2 + y^2). \end{aligned}$$

An *affine plane conic* is the zero set of a polynomial of degree two in two variables, and thus, the six varieties above are all examples of plane conics.

- (d) Which of the defining polynomials in the previous part are irreducible over $\mathbb{R}$, but reducible over $\mathbb{C}$? For these examples, how do the varieties over $\mathbb{R}$ and $\mathbb{C}$ differ?
- (e) Prove that for any polynomials $f, g \in \mathbb{K}[x_1, \dots, x_n]$ the following identity holds:

$$\mathbb{V}(fg) = \mathbb{V}(f) \cup \mathbb{V}(g).$$

Thus factorization is an algebraic signal that the corresponding variety can be broken into (simpler) pieces.

- (f) A nonempty algebraic variety $X \subset \mathbb{K}^n$ is *reducible* if and only if there exists proper algebraic varieties $X_1, X_2 \subset \mathbb{K}^n$, with $X_i \neq X$ for $i = 1, 2$, such that $X = X_1 \cup X_2$. A variety is

irreducible if it is not reducible. Prove that a nonempty algebraic variety $X \subset \mathbb{K}^n$ is irreducible if and only if given proper algebraic varieties $X_1, X_2 \subset \mathbb{K}^n$ such that $X = X_1 \cup X_2$ then $X_i = X$ for some i . (Note here proper means a proper nonempty subset of $\mathbb{K}^n$.)

(g) Note being reducible or irreducible very much depends on the ground field we are working over. For example, prove the following:

- (i) The variety $\mathbb{V}_{\mathbb{K}}(xy)$ is reducible over all fields $\mathbb{K}$.
- (ii) The variety $\mathbb{V}_{\mathbb{K}}(x^2 - y^2)$ is reducible if and only if $\mathbb{K}$ does not have characteristic two. What happens when $\mathbb{K}$ is characteristic two?
- (iii) The variety $\mathbb{V}_{\mathbb{K}}(x^2 + y^2)$ is irreducible when $\mathbb{K} = \mathbb{Q}$ or $\mathbb{R}$, but is reducible when $\mathbb{K} = \mathbb{C}$.

THE NULLSTELLENSATZ

The same algebraic set can be presented by many different ideals. For example, over every field we have that $\mathbb{V}(x) = \mathbb{V}(x^2) = \mathbb{V}(x^3)$. Geometrically, the equations $x = 0$, $x^2 = 0$, and $x^3 = 0$ all impose the same conditions on a point. Algebraically, however, the ideals $\langle x \rangle$, $\langle x^2 \rangle$, and $\langle x^3 \rangle$ are all different. (Be sure you know why.) More generally, if $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ is an ideal and $f^k \in I$ for some $k \in \mathbb{Z}_{\geq 1}$, then for any point $p \in \mathbb{V}(I)$ we have

$$(f(p))^k = f^k(p) = 0.$$

Since $\mathbb{K}$ is a field, it has no nonzero nilpotents, so this forces $f(p) = 0$. Thus every equation whose power belongs to I already vanishes on $\mathbb{V}(I)$, even if the equation itself does not belong to I . This leads to the following operation.

Definition 2. Let R be a commutative ring. The *radical* of an ideal $I \subseteq R$ is

$$\sqrt{I} := \{f \in R \mid f^k \in I \text{ for some } k \geq 1\}.$$

An ideal is *radical* if $I = \sqrt{I}$.

As argued above, and as you will formalize shortly, $\mathbb{V}(I)$ and $\mathbb{V}(\sqrt{I})$ are equal. Hence, taking radicals is one explanation for why the operation $I \mapsto \mathbb{V}(I)$ loses at least some information: it cannot distinguish an ideal from its radical. Amazingly, when $\mathbb{K}$ is algebraically closed, this is the only information that is lost in the process. This is the first of a series of results broadly known as ‘Hilbert’s Nullstellensatz’, each of which highlights a different perspective of what it might mean to “solve” a system of polynomial equations.

Theorem 3 (Hilbert’s Nullstellensatz I (Geometric Form)). *Let $\mathbb{K}$ be an algebraically closed field. The map $I \mapsto \mathbb{V}(I)$ gives a bijection:*

$$\left\{ \begin{array}{l} \text{radical ideals} \\ \text{in } \mathbb{K}[x_1, \dots, x_n] \end{array} \right\} \xrightarrow{\mathbb{V}} \left\{ \begin{array}{l} \text{algebraic varieties} \\ \text{in } \mathbb{K}^n \end{array} \right\}.$$

It is worth pausing to see how this mirrors the relationship between equations and solution sets recalled at the beginning of this lecture. A complete proof of the Nullstellensatz is somewhat beyond the scope of this course, and so we will not give one here. However, we will frequently make use of it and will try to refine our understanding of it.

The theorem underlying this modern geometric formulation was proved, for homogeneous polynomials, in Section 3 of Hilbert's 1893 paper *Ueber die vollen Invariantensysteme*. It appeared three years after his 1890 paper *Ueber die Theorie der algebraischen Formen*, which contained what we now call Hilbert's Basis Theorem and applied it to the studying the question of when the ring of invariants is finitely generated. Famously, Hilbert's 1890 argument was highly non-constructive, leading his contemporary Paul Gordan to exclaim, "Das ist keine Mathematik, das ist Theologie", "This is not mathematics, this is theology". While the provenance of this remark is a bit shaky, the first record of I could find it in-print is from Max Noether in 1914, Hilbert's 1893 paper also concered the problem of studying invariants, however from a more constructive perspective. Interesting, Gordan subsequently gave his own proof of Hilbert's Basis Theorem, and it is his approach we shall roughly follow in Lecture 3.

(4) **Radical Ideals.** Let R be a ring and $I \subseteq R$ an ideal.

- (a) Show that $I \subseteq \sqrt{I}$ and that $\sqrt{\sqrt{I}} = \sqrt{I}$.
 - (b) Suppose $f, g \in \sqrt{I}$, with $f^r \in I$ and $g^s \in I$. Use the binomial theorem to show that $(f + g)^{r+s} \in I$. Deduce that $\sqrt{I}$ is closed under addition.
 - (c) Show that if $f \in \sqrt{I}$ and $h \in R$, then $hf \in \sqrt{I}$. Conclude that $\sqrt{I}$ is an ideal.
 - (d) Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal. Prove that $\mathbb{V}(I) = \mathbb{V}(\sqrt{I})$. In other words, passing to the radical does not change its zero set.
 - (e) Compute $\sqrt{\langle x^3y^2 \rangle} \subseteq \mathbb{K}[x, y]$. Compare the zero sets of $\langle x^3y^2 \rangle$ and its radical.
 - (f) Give three different ideals in $\mathbb{K}[x, y]$ which define the x -axis. Which, if any, of your ideals are radical?
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The geometric form of Hilbert's Nullstellensatz I asserts that $\mathbb{V}$ has an inverse, but it does not yet tell us what that inverse is. This returns us to the inverse problem with which we began: given a set of points, which polynomial equations hold on all of them? The natural answer is to collect every such equation.

Definition 4. For any subset $X \subseteq \mathbb{K}^n$, its *vanishing ideal* is

$$\mathbb{I}(X) := \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(p) = 0 \text{ for every } p \in X\}.$$

Thus $\mathbb{I}(X)$ records every polynomial vanishing on all of X , rather than one particular list of equations describing X . Notice this definition makes sense whether or not X is a variety. When X is a variety we often call $\mathbb{I}(X)$ the *defining ideal* of X . As we shall see shortly, it only really makes sense to think of $\mathbb{I}(X)$ as defining X when X is itself an algebraic variety.

In general the operations $\mathbb{V}$ and $\mathbb{I}$ are not literal inverses, even when we restrict our attention to only algebraic varieties. They reverse inclusions and satisfy

$$X \subseteq \mathbb{V}(\mathbb{I}(X)) \quad \text{and} \quad I \subseteq \mathbb{I}(\mathbb{V}(I)).$$

The first enlargement replaces an arbitrary set by the smallest algebraic set containing it; the second enlarges an ideal to include every polynomial which vanishes on its zero set. Exercise 5 asks you to make the first statement precise.

Two words of warning are useful at this moment: i) A finite list of equations whose common zero set is X need not generate the full ideal $\mathbb{I}(X)$. ii) A priori the ideal $\mathbb{I}(X)$ need not have a finite generating set. This claim is the content of Hilbert's Basis Theorem.

(5) The Inverse Problem & Zariski Closure. Let $X, Y \subseteq \mathbb{K}^n$ be arbitrary subsets.

- (a) Prove that $\mathbb{I}(X)$ is a radical ideal. Hint: First check it is an ideal and then check that it is equal to its radical.
- (b) Prove that if $X \subseteq Y$, then $\mathbb{I}(Y) \subseteq \mathbb{I}(X)$.
- (c) Show that $X \subseteq \mathbb{V}(\mathbb{I}(X))$.
- (d) Show that $\mathbb{V}(\mathbb{I}(X))$ is the smallest algebraic subset of $\mathbb{K}^n$ containing X : if $X \subseteq Z$ and Z is algebraic, then $\mathbb{V}(\mathbb{I}(X)) \subseteq Z$. We call this set the *Zariski closure* of X .
- (e) Conclude that X is an algebraic set if and only if $X = \mathbb{V}(\mathbb{I}(X))$.
- (f) Assume now that $\mathbb{K}$ is infinite and let $X = \{(t, t^2) \mid t \in \mathbb{K}^\times\} \subseteq \mathbb{K}^2$. If $f \in \mathbb{I}(X)$, consider the one-variable polynomial $f(t, t^2)$. Use the root bound recalled in Exercise 1(g) to show that it is identically zero, and hence that $f(0, 0) = 0$. Conclude that $(0, 0) \in \mathbb{V}(\mathbb{I}(X))$ even though $(0, 0) \notin X$. In particular, a parabola with one point removed is not an algebraic set.
- (g) In the previous part, where does the assumption that $\mathbb{K}$ is infinite enter? Over the finite field $\mathbb{F}_q$, set $X_q = \{(t, t^2) \mid t \in \mathbb{F}_q^\times\}$. Show that $x^{q-1} - 1 \in \mathbb{I}(X_q)$ but does not vanish at $(0, 0)$, so $(0, 0) \notin \mathbb{V}(\mathbb{I}(X_q))$. Also explain why $t^q - t$ is a nonzero polynomial in $\mathbb{F}_q[t]$ even though it vanishes at every element of $\mathbb{F}_q$. Identify the inference from part (f) which fails over a finite field.

Assume now that $\mathbb{K}$ is algebraically closed. Exercises 4 and 5 allow us to identify the inverse promised by Hilbert's Nullstellensatz I. Given $I \subseteq \mathbb{K}[x_1, \dots, x_n]$, the ideals $\sqrt{I}$ and $\mathbb{I}(\mathbb{V}(I))$ are both

radical, and those exercises show that they have the same zero set:

$$\mathbb{V}(\sqrt{I}) = \mathbb{V}(I) = \mathbb{V}(\mathbb{I}(\mathbb{V}(I))).$$

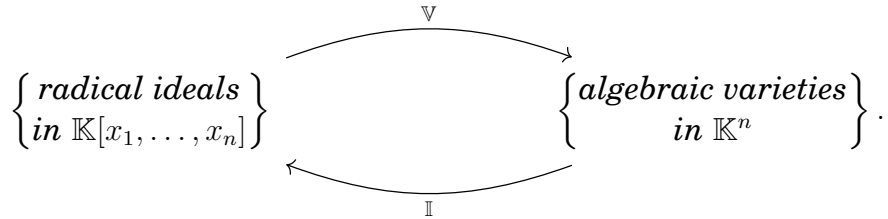
The injectivity in Theorem 3 therefore forces them to be equal. The set $\mathbb{V}(\mathbb{I}(X))$ is the smallest algebraic set containing X ; in particular, if X is already algebraic, then $\mathbb{V}(\mathbb{I}(X)) = X$. The other composition, $\mathbb{I}(\mathbb{V}(I))$, contains the deeper content of the Nullstellensatz, and the equality $\mathbb{I}(\mathbb{V}(I)) = \sqrt{I}$ is false over a general field. Under our algebraically closed hypothesis, this brings us to our second version of the Nullstellensatz.

Theorem 5 (Hilbert's Nullstellensatz II (Recovery Form)). *Let $\mathbb{K}$ be algebraically closed and let $\mathcal{S} \subseteq \mathbb{K}[x_1, \dots, x_n]$. If $V = \mathbb{V}(\mathcal{S}) \subseteq \mathbb{K}^n$, then*

$$\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}.$$

An immediate consequence of Hilbert's Nullstellensatz II is that if I is radical, then $\mathbb{I}(\mathbb{V}(I)) = I$. Thus the vanishing-ideal operation $\mathbb{I}$ is precisely the inverse promised by the bijection $I \mapsto \mathbb{V}(I)$ in Hilbert's Nullstellensatz I. Let us record this conclusion one final time, as this will be the most memorable and used form of the Nullstellensatz.

Theorem 6 (Hilbert's Nullstellensatz III (Affine Algebra–Geometry Dictionary)). *Let $\mathbb{K}$ be an algebraically closed field. The maps $\mathbb{V}$ and $\mathbb{I}$ induce inverse, inclusion-reversing bijections:*



In many ways much of algebraic geometry can be viewed as understanding how various properties can and cannot be passed back and forth thru these correspondences. In this form, the Nullstellensatz solves the inverse problem completely: over an algebraically closed field, an algebraic set determines its radical ideal of equations, and that ideal determines the algebraic set.

A particularly important special case asks which ideal corresponds to the empty set. Since every polynomial vanishes vacuously on the empty set, $\mathbb{I}(\emptyset) = \mathbb{K}[x_1, \dots, x_n]$. Hilbert's Nullstellensatz II therefore gives the following.

Theorem 7 (Hilbert's Weak Nullstellensatz). *If $\mathbb{K}$ is algebraically closed and $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ is an ideal, then $\mathbb{V}(I) = \emptyset$ if and only if $I = \langle 1 \rangle$.*

Notice the reverse implication that $1 \in I$ implies $\mathbb{V}(I) = \emptyset$ is immediate since as a polynomial 1 never vanishes. This is true over any field since in a field $1 \neq 0$. Hence the difficult part of the Weak Nullstellensatz is the forward implication that if $\mathbb{V}(I) = \emptyset$ then $1 \in I$. This implication is where the hypothesis that $\mathbb{K}$ be algebraically closed is crucial. (Consider $\mathbb{V}_{\mathbb{R}}(x^2 + 1)$.)

Although we have obtained the weak form here as a consequence of Hilbert's Nullstellensatz II, the usual proof proceeds in the opposite direction. One first proves the weak Nullstellensatz and then uses Rabinowitsch's trick to deduce the full recovery statement $\mathbb{I}(\mathbb{V}(I)) = \sqrt{I}$. While we shall see Rabinowitsch's trick shortly, it will be of interest to us mainly for its computational consequences.

As an interesting historical aside, Rabinowitsch's trick appeared in a one-page note by J. L. Rabinowitsch. The note, *Zum Hilbertschen Nullstellensatz*, was received in 1929 and published on page 520 of volume 102 of *Mathematische Annalen* in 1930. The page identifies its author only by the initials J. L. and the location Moscow. A colorful story, often repeated to identify J. L. Rabinowitsch with George Yuri Rainich, goes as follows. Rainich supposedly used the trick during a lecture and was interrupted by someone who indignantly accused him of presenting the famous Rabinowitsch trick as his own. Without replying, Rainich turned to the blackboard, wrote RABINOWITSCH, erased the letters *B, O, W, T, S* to leave RAINICH, and then continued his lecture. It is a wonderful story, but almost certainly not history. The current best evidence points towards a different Moscow based mathematician: Yuli Lazarevich Rabinovich, about whom little seems to be known.

(6) Consequences and Certificates. Assume $\mathbb{K}$ is algebraically closed unless otherwise stated.

- (a) Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal. Use Theorem 5 to prove that a polynomial h vanishes at every point of $\mathbb{V}(I)$ if and only if $h^k \in I$ for some $k \geq 1$.
- (b) Let $I = \langle x, xy - 1 \rangle \subseteq \mathbb{K}[x, y]$. Find an explicit identity expressing 1 as a polynomial combination of the displayed generators. Explain what this proves about $\mathbb{V}(I)$.
- (c) Why does the fact that $\mathbb{V}_{\mathbb{R}}(x^2 + 1) = \emptyset$ not contradict Hilbert's Nullstellensatz?
- (d) Return to the circle–parabola system in Exercise 1 given by $f_1 = x^2 - y$ and $f_2 = x^2 + y^2 - 1$. Prove that the polynomial $h = y^2 + y - 1$ vanishes on the common zeros of $f_1 = f_2 = 0$ by explicitly demonstrating that $h \in \langle f_1, f_2 \rangle$. Hint: Express h in terms of f_1 and f_2 .
- (e) *Further: Rabinowitsch's trick.* Let $R = \mathbb{K}[x_1, \dots, x_n]$ and let $I \subseteq R$ be an ideal. Suppose $g \in R$ vanishes on $\mathbb{V}(I)$. Introducing a new variable z consider the polynomial ring $R[z] \cong \mathbb{K}[x_1, \dots, x_n, z]$ and set $J = \langle I, 1 - zg \rangle \subset R[z]$. Notice that variety $\mathbb{V}(J)$ is a subset of $\mathbb{K}^{n+1}$.
 - (i) Prove that $\mathbb{V}(J) = \emptyset$. Use the weak Nullstellensatz to conclude that $1 \in J$.
 - (ii) More generally, prove that $g \in \sqrt{I}$ then $1 \in \langle I, 1 - zg \rangle$. (Hint: If $g^k \in I$ use the geometric series identity for $1 - (zg)^k$.)

The converse is also true, however, it is much more difficult to prove. Put together we get what is known as Rabinowitsch's trick: $g \in \sqrt{I}$ if and only if $1 \in \langle I, 1 - zg \rangle$. We shall use this statement in Lecture 3 to turn a membership test for the ideal on the right hand side into an algorithm to check membership in the radical ideal on the left hand side.

3. PARAMETRIZATIONS AND KERNELS

Up to this point, we have mostly focused on describing a solution set (i.e. a variety) implicitly, through its defining equations. A second perspective to describing a variety is by parametrizing it. Roughly, one should think of a parametrization as a convenient recipe for producing points of the solution set without having to solve the system of equations defining it. For example, the parabola from the beginning of the lecture can be described implicitly as $\mathbb{V}(y-x^2)$, but it can also be described parametrically as the set $\{(t, t^2) \mid t \in \mathbb{K}\}$. These two descriptions are useful for different purposes: the implicit equation $y - x^2$ makes it exceptionally easy to test whether a point $p \in \mathbb{K}^2$ lies on the parabola, while the parametric description makes it easy to construct points on the curve.

More generally, a collection of polynomials $f_1, \dots, f_n \in \mathbb{K}[t_1, \dots, t_m]$ determines a polynomial map

$$F : \mathbb{K}^m \longrightarrow \mathbb{K}^n \quad \text{given by} \quad a \longmapsto (f_1(a), \dots, f_n(a)).$$

We call this a *polynomial parametrization* of its image. Note the parametrization F need not be injective, and the image of F need not be algebraic, i.e., the image might not be a variety). The process of finding the vanishing ideal of the image of F from the polynomials $f_1, \dots, f_n$ defining F is called *implicitization*. More generally, we might define rational parameterizations by replacing the $f_1, \dots, f_n$ by rational functions. Understanding which varieties can be rationally parameterized, and when such a parametrization is generically one-to-one, has been of great interest in algebraic geometry since the 19th century. These questions lead us to the notions of unirational and rational varieties, and more generally birational geometry, which we shall see in Lecture 4.

For the moment, we wish to see how algebra translates the map F into equations. Notice that we may use the same polynomials $g_1, \dots, g_n$ defining F to construct a ring homomorphism

$$\begin{aligned} F^* : \mathbb{K}[x_1, \dots, x_n] &\longrightarrow \mathbb{K}[t_1, \dots, t_m], \\ f &\longmapsto f(g_1, \dots, g_n), \end{aligned}$$

which we call the *substitution map* for $g_1, \dots, g_n$. A polynomial f belongs to $\ker(F^*)$ exactly when $f(g_1, \dots, g_n)$ is the zero polynomial in $\mathbb{K}[t_1, \dots, t_m]$. Any such f certainly vanishes at every point of the image, since $f(F(a)) = (F^*f)(a) = 0$ for all $a \in \mathbb{K}^m$. Hence we always have

$$\begin{aligned} \ker(F^*) &\subseteq \mathbb{I}(F(\mathbb{K}^m)), \\ F(\mathbb{K}^m) &\subseteq \mathbb{V}(\ker(F^*)). \end{aligned}$$

The relationship among these sets depends both on the field and on whether the image is algebraic. Suppose $\mathbb{K}$ is infinite. A polynomial in $\mathbb{K}[t_1, \dots, t_m]$ which vanishes at every point of $\mathbb{K}^m$ is the zero polynomial; this follows by induction on m from the corresponding fact for one-variable polynomials. Consequently, $f \in \mathbb{I}(F(\mathbb{K}^m))$ forces $F^*(f) = 0$, and therefore $\ker(F^*) = \mathbb{I}(F(\mathbb{K}^m))$. In this case, $\mathbb{V}(\ker F^*) = \mathbb{V}(\mathbb{I}(F(\mathbb{K}^m)))$ is precisely the Zariski closure of the parametrized image, and it cuts out the image itself when that image is already algebraic. For a general field, $\ker(F^*)$ can be strictly smaller than $\mathbb{I}(F(\mathbb{K}^m))$, as we shall see in the exercise below.

There are several closely related but distinct problems concerning a parameterization F , which are worth keeping separate in our minds:

- (i) Deciding whether the image of F is algebraic.
- (ii) Computing the kernel of the substitution map F^* .
- (iii) Computing the vanishing ideal of $F(\mathbb{K}^m)$, i.e., computing $\mathbb{I}(F(\mathbb{K}^m))$.

Answering (i) means proving there exists an ideal $J \subset \mathbb{K}[x_1, \dots, x_n]$ such that $\mathbb{V}(J) = F(\mathbb{K}^m)$. If we find such an ideal J it need not be equal to $\mathbb{I}(F(\mathbb{K}^m))$, instead over an arbitrary field all we know is that $J \subset \mathbb{I}(F(\mathbb{K}^m))$. Moreover, although when $\mathbb{K}$ is infinite we have that $\ker(F^*) = \mathbb{I}(F(\mathbb{K}^m))$, this can fail over a finite field. Further, this abstract equality is not yet an explicit answer to the problem of implicitization, as we need a way to find explicit generators for $\ker(F^*)$. At the moment, it is not even clear that such a *finite* generating set exists, yet alone how we might construct one. This will be the focus of Lecture 3 where we will see Gröbner bases and Hilbert's Basis Theorem. The following exercise shows how the three questions above can differ in a relatively simple example.

(7) From a Parametrization to Equations: The Affine Twisted Cubic. Consider the polynomial map

$$F : \mathbb{K} \longrightarrow \mathbb{K}^3 \quad \text{given by} \quad t \longmapsto (t, t^2, t^3),$$

and let C denote the image of F . This curve is called the *affine twisted cubic*; the reason for the name will become clearer when we introduce projective space. Substitution gives the ring homomorphism

$$\begin{array}{ccc} \mathbb{K}[x, y, z] & \xrightarrow{F^*} & \mathbb{K}[t] \\ x & \longmapsto & t \\ y & \longmapsto & t^2 \\ z & \longmapsto & t^3 \end{array}$$

which is the identity on $\mathbb{K}$.

- (a) Verify that $y - x^2$ and $z - xy$ belong to $\ker(F^*)$. Let $J = \langle y - x^2, z - xy \rangle$. Explain why this proves that $J \subseteq \ker(F^*)$ and $C \subseteq \mathbb{V}(J)$.
- (b) Conversely, let $(a, b, c) \in \mathbb{V}(J)$. Show that $b = a^2$ and $c = a^3$, and hence that $(a, b, c) = F(a)$.
- (c) Conclude that $C = \mathbb{V}(J)$. Hence, in this example, the image of the parametrization is already algebraic. This equality holds over every field.
- (d) At this point we know that $J \subseteq \ker(F^*)$ and

$$\mathbb{V}_{\mathbb{K}}(J) = \mathbb{V}_{\mathbb{K}}(\ker(F^*)) = C.$$

However, as we have seen two ideals defining the same variety does not imply the ideals are equal. Further, over an arbitrary field $\mathbb{V}(I) = \mathbb{V}(J)$ does not imply that $\sqrt{I} = \sqrt{J}$. (What assumption are we missing?) The remainder of the exercise proves $\ker(F^*) = J$. For $\alpha, \beta, \gamma \in \mathbb{Z}_{\geq 0}$, show that

$$x^\alpha y^\beta z^\gamma \equiv x^{\alpha+2\beta+3\gamma} \pmod{J}.$$

Hint: First use the generators of J to show that $y \equiv x^2$ and $z \equiv x^3 \pmod{J}$.

- (e) Deduce that every polynomial in $\mathbb{K}[x, y, z]$ is congruent modulo J to a polynomial in x alone.
- (f) Since $J \subseteq \ker(F^*)$, the map F^* factors as follows:

$$\begin{array}{ccc} \mathbb{K}[x, y, z] & \xrightarrow{F^*} & \mathbb{K}[t] \\ \downarrow \pi & \nearrow \overline{F^*} & \\ \mathbb{K}[x, y, z]/J & & \end{array}$$

where π is the natural quotient map. Show that $\overline{F^*}$ is surjective.

- (g) Prove that $\overline{F^*}$ is injective. Then apply the first isomorphism theorem to conclude that $\ker(F^*) = J$. *Hint:* Represent a class in the quotient $\mathbb{K}[x, y, z]/J$ by a polynomial $r(x) \in \mathbb{K}[x, y, z]$ only involving x . Using part (e), then show that $\overline{F^*}(r(x)) = r(t)$.
- (h) We always have $\ker(F^*) \subseteq \mathbb{I}(C)$. The reverse inclusion requires more care. Suppose that $\mathbb{K}$ is infinite, and let $h \in \mathbb{I}(C)$. Show that $h(t, t^2, t^3) \in \mathbb{K}[t]$ vanishes at every $t \in \mathbb{K}$, and hence is the zero polynomial. Conclude that $\mathbb{I}(C) = \ker(F^*) = J$.
- (i) Explain why the preceding conclusion requires an infinite field. If $\mathbb{K} = \mathbb{F}_q$, show that $x^q - x \in \mathbb{I}(C)$ even though $F^*(x^q - x) = t^q - t \neq 0$ as an element of $\mathbb{F}_q[t]$; the latter is a nonzero polynomial which vanishes as a function on $\mathbb{F}_q$. Thus $\ker(F^*) \subsetneq \mathbb{I}(C)$.
- (j) Show that $\langle y - x^2, z - xy \rangle$ is equal to $\langle y - x^2, z - x^3 \rangle$. What does this tell us about the nonuniqueness of a chosen list of equations, even though the ideal of all polynomial relations is uniquely determined?
- (k) Verify the further relation

$$xz - y^2 = y(x^2 - y) - x(xy - z) \in J.$$

Record it for now. Lecture 2 will reveal why the two displayed generators do not yet form the kind of organized generating set needed for systematic computation.

As we saw in the previous exercise, the affine twisted cubic admits a polynomial parametrization in a single parameter. Not every curve admits such a parametrization. As a preview of Lecture 4,

consider the affine cubic curve

$$\mathbb{V}_{\mathbb{C}}(y^2 - x^3 + x) \subset \mathbb{K}^2,$$

whose projective closure is the elliptic curve $\mathbb{V}_{\mathbb{P}^2}(y^2z = x^3 - xz^2)$. (Projective space and projective closure will appear in Lecture 3.) We shall see that unlike the twisted cubic, this curve cannot be rationally parameterized.

We can now carry out the same computations in *Macaulay2* (M2), an open-source computer algebra system created by Daniel Grayson and Michael Stillman for algebraic geometry and commutative algebra. It grew out of its predecessor *Macaulay*, written by Dave Bayer and Stillman; David Eisenbud has been a longtime collaborator, and a broad international community now develops the system and its packages. We will use *Macaulay2* through the browser, so nothing needs to be installed. Open either server on the official

[“Try it out” page](#).

In *Macaulay2* one defines something with the `=` symbol. During a session all defined variables, rings, polynomials, ideals, etc. are remembered. In order to clear these things from memory we use the command `restart`, which starts a fresh session. The *Macaulay2* syntax for multiplication is `*`, powers are written with `^`, and addition and subtraction are `+` and `-` respectively. One can suppress the output of a line by placing a semicolon `;` at the end of the line. Once ran all inputs and outputs `i1`, `o1`, and so on; these labels are not part of the commands. For help with a command, one can enter, for example, `help ideal`.

Since computers cannot store infinite decimal expansions, computations with real or complex numbers generally require floating-point approximations. Although dealing with these numerical issues is itself a beautiful area of mathematics, we will avoid them completely. Instead, we will generally compute over the rational numbers, denoted `QQ` in *Macaulay2*, or over a finite field, where arithmetic is exact. Since our equations have rational coefficients, one may show that any polynomial identity proved over $\mathbb{Q}$ remains true if we view the same identity over $\mathbb{C}$ (The fancy way to say this is “after extending scalars to $\mathbb{C}$.”) So for example, a polynomial that factors over $\mathbb{Q}$ also factors over $\mathbb{C}$. However the converse is not true, and identities proven over $\mathbb{C}$ may fail to be true over $\mathbb{Q}$. In general it is good to develop a sense of what precisely a computation done with *Macaulay2* shows, and then think about what theoretical results can be used to generalize it.

One beauty of *Macaulay2* is that much of its syntax mimics mathematical notation; what you type as code will often look like what you would write on the blackboard. For example, `QQ` denotes $\mathbb{Q}$, `ZZ` denotes $\mathbb{Z}$, and, when p is prime, `ZZ/p` denotes $\mathbb{Z}/p\mathbb{Z}$. When q is a prime power, the finite field $\mathbb{F}_q$ is constructed with `GF(q)`. As noted above, there are also `RR` and `CC`, but they use finite-precision approximations rather than exact copies of $\mathbb{R}$ and $\mathbb{C}$. A polynomial ring is specified by listing the coefficient ring followed by variables, separated by commas, in brackets `[]`. For example, the command `R = QQ[x,y,z]` creates the polynomial ring $\mathbb{Q}[x, y, z]$ and gives it the name `R`.

For now, we will treat commands such as `ker` and `eliminate` as black boxes whose outputs we can interpret. Lectures 2 and 3 will develop the algorithms that make these commands possible; both rely on Gröbner basis computations.

The namesake of both *Macaulay2* and the original *Macaulay* is Francis Sowerby Macaulay (1862–1937), an early pioneer in commutative algebra. Macaulay’s career was filled with contrasts. He spent most of it as a schoolmaster at St Paul’s School in London, yet was elected a fellow of the Royal Society in 1928. His work received little attention from many contemporary algebraic geometers in England but was appreciated by Emmy Noether and those around her who were building the modern language of commutative algebra. His 1916 tract *The Algebraic Theory of Modular Systems* likewise received little attention after his death but became highly influential in the second half of the twentieth century. In Macaulay’s terminology, “modular systems” are what we now call ideals and modules. Much of the book is recognizably computational—elimination, resultants, and explicit systems of equations. The name of the software is therefore apt: as Eisenbud and Gray put it, it “recognizes that much of his work is based on examples created by refined computation.” –

- (8) **First Macaulay2 Session.** Let us return to the circle–parabola equations, which is defined by the ideal $I = \langle x^2 - y, x^2 + y^2 - 1 \rangle \subset \mathbb{K}[x, y]$ Enter the commands one line at a time and compare every output with Exercise 1.

```
restart
R = QQ[x,y];
f1 = x^2-y;
f2 = x^2+y^2-1;
I = ideal(f1,f2); -- encode the intersection as an ideal

R
gens I
f2-f1 -- eliminate x
```

The command `R = QQ[x,y]` creates the polynomial ring $R = \mathbb{Q}[x, y]$. After running this line, we can use the variables x and y to construct polynomials in R . For example `f1 = x^2-y` constructs the polynomial $f_1 = x^2 - y$ and *Macaulay2* knows this is a polynomial in R . The command `I = ideal(f1,f2)` creates the ideal $I = \langle f_1, f_2 \rangle \subset R$. In order to see the generators of I we use the command `gens I`, which displays the chosen generators as a row matrix.

- (a) After running the above code explain the three displayed outputs.
- (b) Write code that determines whether $y^2 + y - 1$ lies in the ideal I . What does this mean geometrically? *Hint:* One direct test is `isSubset(ideal(y^2+y-1), I)`, but can you find a more direct method?
- (9) **A Second Macaulay2 Session.** Now consider the parametrization of the affine twisted cubic. Before running the block, predict the two zero outputs and at least one generator of the kernel.

```

restart
A = QQ[x,y,z];
T = QQ[t];
phi = map(T,A,{t,t^2,t^3}); -- encode t |-> (t,t^2,t^3)
phi(y-x^2)
phi(z-x*y)
K = ker phi; -- compute the full ideal of relations
trim K

```

Here `ker phi` computes an ideal, while `trim K` asks for a smaller list of generators.

- Macaulay2* lists the target of a ring map first and its source second. Explain why the line `map(T,A,{t,t^2,t^3})` above represents the map $A \rightarrow T$ used in Exercise 7.
- Explain the two zero outputs. Compare the generators in `trim K` with the generators found by hand. Why do the displayed lists not need to agree term for term?
- Use `K == ideal(y-x^2,z-x*y)` to check equality of the two ideals. What mathematical statement is being verified by the output `true`?
- Testing one polynomial at a time is already becoming tedious. A *Macaulay2* list is enclosed in braces. Enter

```

candidates = {y-x^2,z-x*y,x*z-y^2,x};
apply(candidates,g -> phi(g))

```

The arrow expression on that line defines a function, in this case, taking something named g to $\phi(g)$, i.e., evaluating the map ϕ defined above at the element g . The command `apply` applies a function to every element of a list or a sequence before returning the outputs as a new list or new sequence. (Note your original list remains unchanged.) Before running the second line, predict the list it should return.

- The second useful command to know is `scan`, which also applies a function to every entry of a list or sequence. However, `scan` discards the returned values, as such it useful when the function itself returns an output, such as printing. Run the following code.

```

scan(candidates,g -> print (g,phi(g) == 0))

```

What is the meaning of the output? How does the differ from the output you got when you ran the `apply` in the previous part?

- Add polynomial polynomial of your choosing to the list `candidates`, predict whether it belongs to $\ker(\phi)$, and use the same commands to test your prediction.

- (g) *Further:* Also verify membership in K using `(x*z-y^2) % gb K`. Explain the mathematical meaning of each test.
-

In addition to being open source, much of the specialized functionality of *Macaulay2* is developed by researchers in commutative algebra and algebraic geometry through *packages*: collections of commands and documentation devoted to a particular area of mathematics or type of computation. These packages are written and maintained by members of the *Macaulay2* community, allowing the program to grow alongside current research. A list of the packages distributed with *Macaulay2*, organized by subject, is available in the [package index](#).

Before using a package, we load it with the command `needsPackage`. For example, the command `needsPackage "Elimination"` loads the package `Elimination`. Loading a package makes all of the commands in the package available for the current session, however, we must load the package each time we restart if we wish to continue using it. Every *Macaulay2* package comes with extensive documentation, which includes an overview of the package, a list of functions, methods, and objects, and a host of examples. You access the package documentation via the command `viewHelp`, for example, `viewHelp "Elimination"`, which will open the documentation as an HTML file. I would encourage you to frequently consult and use the *Macaulay2* documentation; one rarely memorizes every command, and it is often important to carefully understand exactly what a command is computing.

In the following exercise we will use the command `dim I`, which computes the Krull dimension of R/I for an ideal $I \subseteq R$. The precise definition of Krull dimension is beyond this course, but you should informally think of it as the number of locally independent parameters of the corresponding affine variety. For example, an ideal defining a non-empty finite set has Krull dimension zero, an ideal defining a curve has Krull dimension 1, and an ideal defining a surface has Krull dimension 2. (Note we define the dimension relative to the ground field $\mathbb{K}$, and so we think of $\mathbb{C}$ as being one-dimensional while $\mathbb{C}^2$ is two-dimensional. Although we should be careful not to confuse Krull dimension with vector space dimension. We will also use the command `degree I`, which computes the degree of R/I . When $\mathbb{V}(I)$ is zero-dimensional, `degree I` is equal to the $\mathbb{Q}$ -vector-space dimension of R/I . As we will see, in this setting, we should roughly think of degree as counting the number of points in $\mathbb{V}(I)$ if we count the points with the correct multiplicity. Exercises 12 and 13 will make this precise by comparing the degree of an ideal defining four distinct points with the degree of an ideal defining a tangential double point.

- (10) **A Third *Macaulay2* Session.** Let us return to the circle–parabola ideal and start a fresh *Macaulay2* session. Before running the block below, predict the two elimination generators, the dimension, and the degree.

```

restart
needsPackage "Elimination";
R = QQ[x,y];
I = ideal(x^2-y,x^2+y^2-1);

eliminate({y},I) -- algebraic equations involving only x
eliminate({x},I) -- algebraic equations involving only y

dim I == 0 -- check that the intersection is a finite set of points
degree I -- count (with multiplicity) the number of points in the
            intersection

```

- (a) Match the two elimination ideals with the hand calculations in Exercise 1. Notice that eliminating y leaves equations in x , and eliminating x leaves equations in y .
- (b) Give a geometric interpretation of the outputs of `dim I` and `degree I`. (Note there is something weird going on here; how many points does $x^2 - y$ and $x^2 + y^2 - 1$ intersect in over $\mathbb{Q}$, which is the field we are working over.)
- (c) *Further: Closure Rather Than Image.* Set $H = \text{ideal}(x*y-1)$ and eliminate y with the following command:

```
eliminate({y},H)
```

Compare the output with Exercise 1(g): which point has elimination added, and why?

- (d) *Further: Components.* The following computation returns to the factorization examples from Exercise 3. Predict the visible components before asking the computer.

```

restart
needsPackage "MinimalPrimes";
R = QQ[x,y];

axes = ideal(x*y);
decompose axes -- the two irreducible components

twoLines = ideal((x-y)*(x+y));
decompose twoLines -- factorization becomes component decomposition

```

Compare the outputs with the decompositions found by hand. In Lecture 5, we will return to primary decomposition and to nonreduced equations, where they have a geometric job.

4. COORDINATE RINGS AND MULTIPLICITY

We now return to a distinction already encountered in the equality $\mathbb{V}(x) = \mathbb{V}(x^2)$. A polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$ restricts to a function $X \rightarrow \mathbb{K}$ on any subset $X \subseteq \mathbb{K}^n$. We say that a function $\phi : X \rightarrow \mathbb{K}$ is a *polynomial function on X* if there exists a polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$ such that $\phi = f|_X$. Note sometimes it is possible for polynomial functions to hide, and not look like polynomials. Consider for example the function $(x, y) \mapsto e^{x^2+y^2}$. This is a well-defined function on all of $\mathbb{R}^2$, but it is certainly not a polynomial function on $\mathbb{R}^2$. However, if we restrict this and consider it as a function on the unit circle, $\mathbb{V}_{\mathbb{R}}(x^2 + y^2 - 1)$, it is a polynomial function. Can you find the polynomial f it agrees with on $\mathbb{V}_{\mathbb{R}}(x^2 + y^2 - 1)$?

Since function addition, subtraction, and multiplication are defined point-wise, the set of all functions $X \rightarrow \mathbb{K}$ forms a ring. (You should think about this carefully if it is not obvious.) Further, more, since restriction commutes with these operations the subset of polynomial functions on X forms a subring. This motivates the following definition.

Definition 8. The *coordinate ring* of an affine algebraic variety $X \subseteq \mathbb{K}^n$, which we denote by $\mathbb{K}[X]$, is the ring of polynomial functions on X under point-wise addition and multiplication.

While the above definition makes it seem very abstract, the coordinate ring of a variety X , has a beautiful and simple description. Namely, by definition there is a surjective map of rings:

$$\mathbb{K}[x_1, \dots, x_n] \xrightarrow{\rho} \mathbb{K}[X] \quad f \longmapsto f|_X$$

where $f|_X$ is the polynomial function on X determined by f , i.e. $p \mapsto f(p)$. A polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$ restricts to give the zero function on X if and only if $f(p) = 0$ for all $p \in X$. Hence a polynomial f is in the kernel of ρ if and only if f vanishing on all of X , i.e. if and only if $f \in \mathbb{I}(X)$. Hence the kernel of ρ is precisely $\mathbb{I}(X)$. Using everyone's favorite isomorphism theorem we get:

$$\mathbb{K}[X] := \mathbb{K}[x_1, \dots, x_n] / \mathbb{I}(X).$$

Since $\mathbb{I}(X)$ is a radical ideal the coordinate ring of X is always reduced, i.e., it has no nonzero nilpotent elements. Further, if $F : X \rightarrow Y$ is a map of varieties that is given in coordinates by polynomials and $\phi : Y \rightarrow \mathbb{K}$ is a polynomial function then the composition $\phi \circ F : X \rightarrow \mathbb{K}$ is a polynomial function. Hence F “pullsback” polynomial functions and induces a ring homomorphism

$$F^* : \mathbb{K}[Y] \longrightarrow \mathbb{K}[X], \quad \text{given by} \quad [h] \longmapsto [h \circ F].$$

Notice this ring homomorphism goes in the opposite direction to the map of algebraic varieties.

Now set $S = \mathbb{K}[x_1, \dots, x_n]$ and suppose that $X = \mathbb{V}(I)$ for a specified ideal $I \subseteq S$. We wish to distinguish the coordinate ring $\mathbb{K}[X]$ from the quotient S/I attached to the ideal with which we began. The inclusion $I \subseteq \mathbb{I}(\mathbb{V}(I))$ induces a canonical surjection

$$S/I \longrightarrow \mathbb{K}[\mathbb{V}(I)] = S/\mathbb{I}(\mathbb{V}(I)).$$

Its kernel is $\mathbb{I}(\mathbb{V}(I))/I$. When $\mathbb{K}$ is algebraically closed, the Nullstellensatz identifies this map with

$$S/I \longrightarrow S/\sqrt{I},$$

whose kernel $\sqrt{I}/I$ consists precisely of the nilpotent elements of S/I . Indeed, the class of f in S/I is nilpotent exactly when $f^m \in I$ for some $m \geq 1$.

This gives a precise meaning to the statement that nilpotents cannot be detected by points. Every $p \in \mathbb{V}(I)$ determines an evaluation homomorphism

$$\text{ev}_p : S/I \longrightarrow \mathbb{K}, \quad \bar{f} \longmapsto f(p).$$

Every nilpotent is sent to zero by every such homomorphism, since a field has no nonzero nilpotents. Conversely, when $\mathbb{K}$ is algebraically closed, the Nullstellensatz says that an element annihilated by every point evaluation must be nilpotent. Thus nilpotents are not additional points hidden from the picture; they are algebraic information retained by S/I but lost when we pass to the underlying solution set. In the finite intersections below, this information records repeated vanishing and tangency.

The algebraically closed hypothesis in the converse is essential. For example, $\langle x^2 + y^2 \rangle$ is radical in $\mathbb{R}[x, y]$, but its real zero set is only the origin, whose vanishing ideal is $\langle x, y \rangle$. Over a non-algebraically closed field, passing to the set of $\mathbb{K}$ -valued solutions may therefore forget more than nilpotents.

(11) Coordinate Rings, Nilpotents, and Points.

Let $S = \mathbb{K}[x_1, \dots, x_n]$.

(a) Let $I \subseteq S$. Prove that S/I is an integral domain if and only if I is prime. Show that the nilpotent elements of S/I are precisely the classes $f + I$ with $f \in \sqrt{I}$. Deduce that S/I is reduced if and only if I is radical.

(b) Let X be a nonempty algebraic set. Using $\mathbb{V}(fg) = \mathbb{V}(f) \cup \mathbb{V}(g)$, prove that

$$\begin{aligned} X \text{ is irreducible} &\iff \mathbb{I}(X) \text{ is prime} \\ &\iff \mathbb{K}[X] \text{ is an integral domain.} \end{aligned}$$

Notice that this equivalence does not require $\mathbb{K}$ to be algebraically closed.

Hint: For the reverse implication, if $X = X_1 \cup X_2$ with both $X_i \subsetneq X$, choose $f_i \in \mathbb{I}(X_i) \setminus \mathbb{I}(X)$.

(c) Let $I \subseteq S$ and suppose that $X = \mathbb{V}(I)$ is nonempty. Show that the natural map $q_I : S/I \longrightarrow \mathbb{K}[X]$ has kernel $\mathbb{I}(X)/I$.

(d) Show that every $p \in X$ induces an evaluation homomorphism $\text{ev}_p : S/I \longrightarrow \mathbb{K}$ and that

$$\ker(q_I) = \bigcap_{p \in X} \ker(\text{ev}_p).$$

(e) Assuming that $\mathbb{K}$ is algebraically closed, use the Nullstellensatz to conclude that this intersection is precisely the set of nilpotent elements of S/I .

(12) **A Double Point from Tangency.** Let $A_1 = \mathbb{K}[x]/\langle x \rangle$ and $A_2 = \mathbb{K}[x]/\langle x^2 \rangle$. Notice that both $\langle x \rangle$ and $\langle x^2 \rangle$ define the single point $\{0\} \subseteq \mathbb{K}$.

(a) Prove that $\bar{x} \neq 0$ in A_2 while $\bar{x}^2 = 0$. Identify the kernel of the natural map $A_2 \longrightarrow A_1 \cong \mathbb{K}[\{0\}]$, and explain why evaluation at the unique point cannot detect $\bar{x}$.

(b) The preceding algebra occurs geometrically when curves meet tangentially. The parabola $y = x^2$ is tangent to the line $y = 0$ at the origin. Set

$$I_{\text{tan}} = \langle x^2 - y, y \rangle \subseteq \mathbb{C}[x, y].$$

Show that $\mathbb{C}[x, y]/I_{\text{tan}} \cong \mathbb{C}[x]/\langle x^2 \rangle$.

(c) Compute $\mathbb{V}(I_{\text{tan}})$, $\sqrt{I_{\text{tan}}}$, and $\mathbb{C}[\mathbb{V}(I_{\text{tan}})]$.

(d) Find a basis for $\mathbb{C}[x, y]/I_{\text{tan}}$ as a $\mathbb{C}$ -vector space and identify its nonzero nilpotent classes.

(e) Explain why the underlying intersection consists of only the origin, while the quotient ring has dimension 2 as a $\mathbb{C}$ -vector space. Since this is the only intersection point, this dimension is its intersection multiplicity: the origin is therefore *counted twice*.

(13) **Four Distinct Intersection Points.** Consider the intersection of the parabola $y = x^2$ with the circle $x^2 + y^2 = 1$, and set

$$I_{\text{circ}} = \langle x^2 - y, x^2 + y^2 - 1 \rangle \subseteq \mathbb{C}[x, y].$$

(a) Show that $\mathbb{C}[x, y]/I_{\text{circ}} \cong \mathbb{C}[x]/\langle x^4 + x^2 - 1 \rangle$.

(b) Deduce that the classes of $1, x, x^2, x^3$ form a $\mathbb{C}$ -basis for this quotient, and use $y = x^2$ in the quotient to represent this basis by the classes of $1, x, y, xy$.

(c) Recall from Exercise 1 that $x^4 + x^2 - 1$ has four distinct complex roots $\alpha_1, \dots, \alpha_4$. Use the Chinese remainder theorem to prove that evaluation induces an isomorphism

$$\mathbb{C}[x]/\langle x^4 + x^2 - 1 \rangle \longrightarrow \mathbb{C}^4, \quad r(x) \longmapsto (r(\alpha_1), \dots, r(\alpha_4)).$$

(d) Conclude that $\mathbb{C}[x, y]/I_{\text{circ}}$ is reduced and that the intersection consists of the four distinct points (α_i, α_i^2) for $i = 1, 2, 3, 4$.

(e) Compare this intersection with the tangent intersection in Exercise 12. Explain precisely why the tangent intersection is described as *one point counted twice*, whereas the circle–parabola intersection consists of *four distinct points*. Your answer should compare the underlying point sets, the vector-space dimensions of the quotient rings, and the presence or absence of nonzero nilpotents.

Exercises 12 and 13 are two halves of one classical statement called Bézout's theorem. Roughly, Bézout's theorem states that two plane curves of degrees d and e with no common component intersect in exactly de points. Thus, a conic and a line intersect in exactly $2 \cdot 1 = 2$ points, while two conics meet in $2 \cdot 2 = 4$ points. Note this can be thought of as a generalization of the fact you might have learned in high school: the degree of a polynomial is equal to the number of intersection points of the graph of the polynomial with a generic horizontal line.

As the previous exercises highlight, in order to make Bézout's theorem true three refinements are needed to how we could intersection points: (i) we must work over an algebraically closed field, (ii) we must count points with multiplicity, and (iii) we must work retain points that intersect "at infinity", which means working in projective space. We will see more about projective space in the later lectures. While this theorem is normally credit to Bézout's work in 1779, the result is also stated, without proof, in Newton's Principia in 1687

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- (14) **The Affine Dictionary.** Assuming $\mathbb{K}$ is algebraically closed, return to the opening table and complete the algebraic column using $1 \in I$, $\sqrt{I}$, elimination ideals, finite quotient dimension, primality of $\sqrt{I}$, and kernels. Fill in the *Macaulay2* entries learned so far: `eliminate`, `dim`, `degree`, and `ker`; if you completed the *Further: Components* part, also include `decompose`. Mark the rows where an output may describe a Zariski closure rather than the exact image. We will return to the remaining entries in the subsequent lectures.